

Table 7-111. Far End Device Type Detection (Sheet 2 of 2)

Device Type	Negotiated Link Direction	Negotiated PBR-Enabled	Negotiated MLD-Enabled	Received "Link Partner Info" Type
GFD	DSP	Y	N	GFD
MLD	DSP	N	Y	N/A
SLD, PCIe EP, or HBR Switch	DSP	N	N	N/A

3. FM may choose to first continue discovery of any connected switches or to manage devices on the far end of all switch ports.

PBR switch PPBs connected as ISLs are configured by the FM with the **Send PPB CXL.io Configuration Request** command.

The FM uses the **Fabric Crawl Out** command, as defined in [Section 7.7.13.2](#), using switch port number as the target to manage the devices on the far end of each switch port. The FM claims ownership and assigns a PID to each defined as covered in step 1.

Once the far end device has been assigned a PID, the FM must program the PBR switch's DRT to enable routing of that PID to the appropriate switch port. The FM can now use this new assigned PID as the target for subsequent **Fabric Crawl Out** requests.

Steps 1 – 3 are repeated for all PBR switches discovered.

7.7.12.3 Assigning and Binding PIDs

As defined in [Section 7.7.6.5](#), there are many entities within a fabric that require PIDs to be assigned. GFDs and PBR switches are assigned a PID for device management purposes when the FM registers with these devices using the **Claim Ownership** command. A PBR switch reports all additional possible PID assignments with the **Get PID Target List** command.

The FM may start performing binding operations after all required PIDs have been assigned using the **Configure PID Assignment** commands. There are two methods for binding, depending on the location of the source and target of the operation. The **Bind vPPB** command is used to bind a direct attached device or LD to a switch's VCS.

The **Configure PID Binding** command is used to bind Downstream ES VCS vUSPs to Host ES vDSPs in a two-step operation. First, a binding command is sent to the Downstream ES, assigning the PID of the Host edge port to a Downstream ES VCS. Assignment of this PID allows the Downstream ES FPorts to select appropriate decoding and routing logic based on the SPID of incoming transactions. As detailed in [Section 7.7.12.4](#), latency and BW values are configured with this binding so that CDAT information can be generated in the Downstream ES.

A binding command is also sent to the Host ES, assigning the PID of the desired Downstream ES FPort and associating the binding with a specified vDSP. The Host ES uses this as the DPID for downstream transactions.

7.7.12.4 Reporting Fabric Route Performance via CDAT

Hosts require CDAT information that defines the attributes and performance characteristics of regions of memory for all memory interconnect configurations, including PBR fabrics. Special mechanisms are defined for determining and reporting this information in a PBR fabric because hosts have no visibility of intermediate ISLs, as outlined in [Section 7.7.6.1](#). The mechanisms used for LD-FAM differ from those used for G-FAM.

7.7.12.4.1 Accessing CDAT Information for LD-FAM

There are up to three components involved in the path to LD-FAM in a PBR fabric: a Host ES, a Downstream ES, and an LD-FAM device. The Host ES and LD-FAM devices require no special handling and report CDAT information covering their own characteristics as they would in an HBR system deployment. The Downstream ES, however, is required to report CDAT information that covers its own device-level performance factoring in the impact of the fabric routing path, as described below.

Latency and BW values are provided when the binding between a Host ES VCS and Downstream ES VCS is configured with the **Configure PID Binding** command. Routes through a fabric are expected to have symmetric performance characteristics. As such, only one latency and BW value is provided to define the fabric routing path. The Downstream ES adds the latency of the routing path to its own latency and uses the lesser of the BW values.

Hosts access CDAT information for Downstream ES VCSs from a DOE instance present in the vUSP.

7.7.12.4.2 Accessing CDAT Information for G-FAM

The access mechanism for CDAT from G-FAM is necessarily different from LD-FAM as a result of 2 key architectural differences: G-FAM is presented through the FAST, not a switch-based topology, and GFDs do not implement nor expose a DOE instance to the host. CDAT access for G-FAM instead relies on the use of CCI opcodes.

The GAE providing G-FAM access is responsible for producing the CDAT for each segment of the FAST. Latency and BW values are provided when PID access is enabled with the **Configure PID Access** command. The CDAT information is queried by the host using the **Read CDAT** command.

GFDs are responsible for providing CDAT information covering their own characteristics. The host queries CDAT information from GFDs using the **Proxy GFD Management** Command request to initiate the **Read CDAT** command.

7.7.12.5 Configuring CacheID in PBR Fabric

From the host's perspective, configuration of CacheID for VFs spanning a PBR Fabric is performed identically to such configuration in an exclusively HBR topology. PBR switches automatically exchange ID configuration information in the following manner:

1. The Downstream ES presents ID route table capabilities in its vPPBs (see [Section 8.2.4.28](#) for details on the CacheID Route Table).
2. The host will enumerate and assign all IDs and program the route table capability, triggering the Commit bit to complete the configuration.
3. The setting of the Commit bit triggers the Downstream ES to generate one or more RTUpdate VDMs, as defined in [Section 3.1.11.7](#), targeted at the Host PID. The Host ES will intercept this VDM based on its PBR opcode.
4. Upon receipt of the VDM, the Host ES programs the necessary ID to PID translation logic in the Host edge port.
5. The Host ES acknowledges successful programming of the ID translation logic with an RTUpdateAck VDM, as defined in [Section 3.1.11.8](#), sent to the Downstream ES for each RTUpdate VDM that was received and successfully processed.
6. Upon receipt of the VDM, the Downstream ES sets the corresponding 'RT Committed' bit in the vUSP.

A downstream HBR switch topology requires PIDs for each unique potential target so that IDs can be translated between CacheID and PID at the fabric edges. For CacheID, the ID is valid if the Valid bit is set in a Cache ID Target entry in the Cache ID Route Table Capability Structure. The corresponding PID used is the PID of the DSP to which the Route Table entry has been configured to map. Multiple PIDs must be assigned to a DSP if multiple IDs map to that DSP.

7.7.12.6 Dynamic Fabric Changes

This section outlines how FMs and PBR switches handle various changes to the system configuration during runtime.

7.7.12.6.1 Hot-Add and Link Up Events

A new Link Up on an unbound edge port is indicated to the FM via a Physical Switch Event Record. The FM uses the **Get Physical Port State** and **Get PBR Link Partner Info** commands to query information on the device connected to the port.

When an SLD or PCIe device is Hot-Added to a bound port, the FM can be notified but is not involved.

7.7.12.6.2 Dynamic Configuration Changes

There are many runtime configuration changes that an FM can trigger on a fabric:

- Binding/Unbinding: New bindings are presented to hosts as hot-add operations. Unbinding an EP is presented as a hot-remove operation.
- Updates to GMV/VTV: The GAE generates a notification to the host when changes are made to the GMV or VTV enabling or disabling access to a particular PID.
- GFD DCD changes: GFDs generate notifications to all impacted GAEs when updates are made to a host group's extent list.

7.7.12.6.3 Hot/Surprise Remove and Link Down Events

The FM is responsible for managing a Link Down event:

- The PBR switch that experienced the Link Down notifies the FM with a Physical Switch Event Record
- EP Link Down events are represented as surprise removes to the host
- The FM manages any required topology changes associated with an ISL Link Down event, including clearing the PID binding between the Upstream ES and Downstream ES VCSs, which is presented to the host as a hot-remove of the Downstream ES VCS
- GFD Link Down events prompt the FM to disable access to the corresponding PID in all impacted hosts' GAE GMV and VTV
- PBR switches drop unroutable transactions

7.7.13 PBR Switch Command Set

This command set is only supported by, and must be supported by, PBR switches to facilitate the discovery of a PBR fabric and configuration of routing and bindings.

7.7.13.1 Identify PBR Switch (Opcode 5700h)

This command provides information to the FM about a PBR switch's fabric capabilities.

Possible Command Return Codes:

- Success
- Unsupported
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-112. Identify PBR Switch Response Payload

Byte Offset	Length in Bytes	Description
0h	8	GAE Support Map: Bitmask indicating whether a VCS includes (1) or does not include (0) a GAE instance in the host edge switch USP or downstream edge switch vUSP where bit position corresponds to VCS ID.
8h	1	Number of DRTs: Total number of DRTs supported by the switch. This value shall be greater than 0.
9h	1	Number of RGTs: Total number of RGTs supported by the switch.
Ah	1	Reserved
Bh	1	• Bit[0]: Random Supported: Indicates whether "Random" dynamic routing mode is supported (1) or not supported (0) • Bit[1]: Congestion Avoidance Supported: Indicates whether "Mix with CA" dynamic routing mode is supported (1) or not supported (0) • Bit[2]: Advanced Congestion Avoidance Supported: Indicates whether "Advanced CA" dynamic routing mode is supported (1) or not supported (0) • Bits[5:3]: Reserved • Bit[6]: Vendor-specific Routing Mode 1 Supported: Indicates whether the vendor-specific routing mode configured by dynamic routing mode value 6 is supported (1) or not supported (0) • Bit[7]: Vendor-specific Routing Mode 2 Supported: Indicates whether the vendor-specific routing mode configured by mode value 7 is supported (1) or not supported (0)

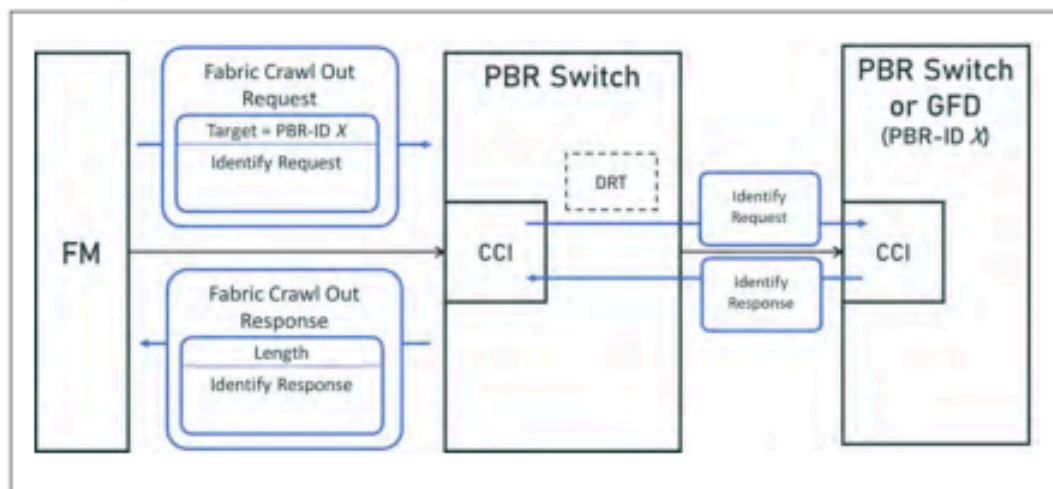
7.7.13.2 Fabric Crawl Out (Opcode 5701h)

This command is used to tunnel management commands at components in a PBR fabric in two scenarios:

- PBR devices with no assigned PID: Tunneled command is sent to the PBR switch to which the PBR device is attached with a target specifying the PBR switch port to which the PBR device is connected. The receiving switch will transmit the command out the specified port using the reserved DPID FFFh.
- PBR devices with an assigned PID: Tunnel command is sent to a PBR switch with a target specifying the PID assigned to the PBR device.

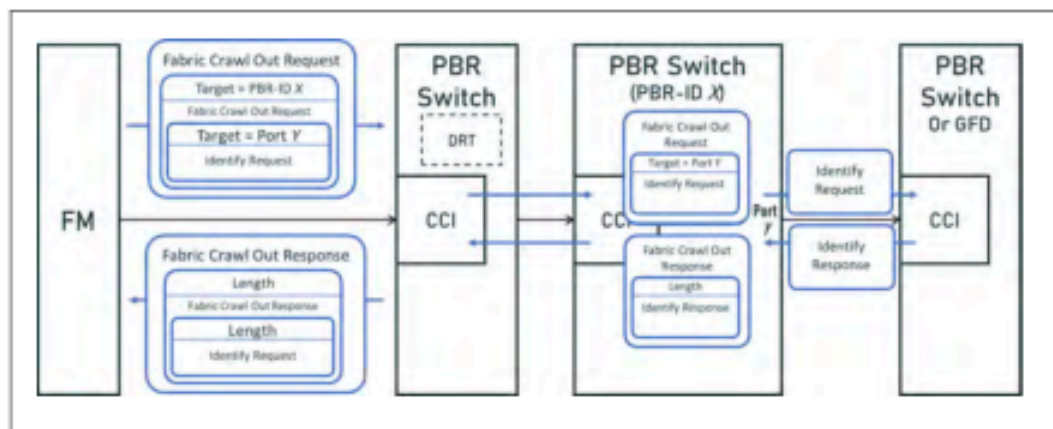
The transport of these commands across PBR links is defined in [Section 7.7.11.6](#).

Figure 7-54. Tunneling Commands to Remote Devices



The Management Command input payload field includes the tunneled command encapsulated in the CCI Message Format, as defined in Figure 7-19. This can include an additional layer of tunneling for commands issued to components with no assigned PID, as illustrated in Figure 7-55.

Figure 7-55. Tunneling Commands to Remote Devices with No Assigned PID



Response size varies, based on the tunneled command's definition. Valid targets for the tunneled commands include PBR switch ports, and PBR devices within a fabric.

This command fails with "Invalid Input" if the target specifies a non-existent switch port or a PID with no valid entry in the DRT.

Components shall terminate the processing of a request that includes more than 2 layers of tunneling and provide an "Unsupported" return code.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error

- Retry Required

Command Effects:

- None

Table 7-113. Fabric Crawl Out Request Payload

Byte Offset	Length in Bytes	Description
0h	2	Target: Encoding depends on Target Type : • Target Type = 0: — Bits[7:0]: Port Number: Switch shall transmit command out specified egress port. — Bits[15:8]: Reserved. • Target Type = 1: — Bits[11:0]: PBR-ID: Target PID. Switch shall determine egress port using DRT. — Bits[15:12]: Reserved. • All other encodings are reserved
2h	1	• Bits[3:0]: Target Type: Specifies the type of tunneling target for this command: — 0h = Port Number: Indicates that the tunneling target is a component on the far end of a switch port — 1h = PBR-ID: Indicates that the tunneling target is a component in the PBR fabric address by a PID — All other encodings are reserved • Bits[7:4]: Reserved
3h	1	Reserved
4h	2	Command Size : Number of valid bytes in Management Command.
6h	Varies	Management Command : Request message formatted in the CCI Message Format as defined in Figure 7-19.

Table 7-114. Fabric Crawl Out Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Response Length : Number of valid bytes in Response Message.
2h	2	Reserved
4h	Varies	Response Message : Response message formatted in the CCI Message Format as defined in Figure 7-19.

7.7.13.3 Get PBR Link Partner Info (Opcode 5702h)

This command reads the data received from the latest “Link Partner Info” VDM on a PBR link.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-115. Get PBR Link Partner Info Request Payload

Byte Offset	Length in Bytes	Description
0h	1h	Number of Ports: Number of ports requested.
1h	Varies	Port ID List: 1-byte ID of requested port, repeated Number of Ports times.

Table 7-116. Get PBR Link Partner Info Response Payload

Byte Offset	Length in Bytes	Description
0h	1	Number of Ports: Number of port information blocks returned.
1h	3	Reserved
4h	Varies	Link Partner Info List: Link Partner Info block as defined in Table 7-117, repeated Number of Ports times.

Table 7-117. Get Link Partner Info Block Format

Byte Offset	Length in Bytes	Description
00h	1	Port ID: Number of the port reporting this Link Partner Info.
01h	1	Far End Port ID: Port number (Port ID) of the source (sender) of the Link Partner Info VDM.
02h	2	- Bits[11:0]: PID: As reported in Link Partner Info VDM - Bits[15:12]: Device Type: As reported in Link Partner Info VDM
04h	10h	Link Partner ID: As reported in Link Partner Info VDM.
14h	1	Standard FC VC List: As reported in Link Partner Info VDM.
15h	1	UIO FC VC List: As reported in Link Partner Info VDM.
16h	10h	Primary FM UUID: As reported in Link Partner Info VDM.
26h	10h	Secondary FM UUID: As reported in Link Partner Info VDM.

7.7.13.4 Get PID Target List (Opcode 5703h)

This command retrieves the list of targets within a PBR switch to which a PID may be assigned. This does not include the PID assigned to the switch itself as part of the **Claim FM Ownership** command. As outlined in Section 7.7.6.5, the following restrictions apply when assigning PIDs:

- A fabric port may be assigned one PID that can be shared among multiple fabric ports
- A Downstream Edge Port may be assigned one PID that must be unique
- A Host Edge Port may be assigned more than one PID, each of which must be unique

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-118. Get PID Target List Request Payload

Byte Offset	Length in Bytes	Description
0h	2	Start Index: Index of first target to return.
2h	2	Number of Targets: Maximum number of targets to return.

Table 7-119. Get PID Target List Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Total Number of Targets: Total number of PID targets supported by the device.
2h	2	Number of Targets: Number of targets returned in Target List .
4h	Varies	Target List: List of PID target as defined in Table 7-120.

Table 7-120. Target List Format

Byte Offset	Length in Bytes	Description
0h	2	Target ID: ID of PID Target for use in Configure PID Assignment.
2h	1	• Bits[2:0]: Target Type: — 000b = Fabric Port — 001b = Host Edge Port (USP/GAE) — 010b = Downstream Edge Port — All other encodings are reserved • Bits[7:3]: Reserved
3h	1	Instance ID: Index of PID for targets that can support multiple PIDs.
4h	1	VCS ID: ID of associated VCS. Valid only when Target Type is 1 (Host Edge Port).
5h	1	Physical Port ID: Physical port ID of the target.
6h	2	• Bits[11:0]: PID: Current PID assignment. FFFh if unassigned. • Bits[15:12]: Reserved.

7.7.13.5 Configure PID Assignment (Opcode 5704h)

This command is used to assign PIDs to targets within a PBR switch.

Note:

This command does not update the corresponding DRT entries for assigned or cleared PIDs. The DRT must be updated separately, using the **Set DRT** command as necessary.

This command shall return Invalid Input under the following conditions:

- Specified target is invalid
- PID has already been assigned to another target within the switch

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error

- Retry Required

Command Effects:

- None

Table 7-121. Configure PID Assignment Request Payload

Byte Offset	Length in Bytes	Description
0h	1	• Bits[2:0]: Operation: Specifies the PID assignment operation: — 000b = Assign PID — 001b = Clear PID — All other encodings are reserved • Bits[7:3]: Reserved
2h	2	Number of Targets : Number of entries in PID Assignment List.
4h	Varies	PID Assignment List : List of PID assignments as defined in Table 7-122.

Table 7-122. PID Assignment

Byte Offset	Length in Bytes	Description
0h	2	• Bits[11:0]: PID: PID to assign to the specified target • Bits[15:12]: Reserved
2h	2	Target ID : Index of PID target, as reported in Get PID Target List response.
4h	1	Instance ID : Index of PID for targets that can support multiple PIDs.

7.7.13.6 Get PID Binding (Opcode 5705h)

This command reads the binding of Downstream ES PIDs to Upstream ES vDSPs or Upstream ES USP PIDs to Downstream ES vUSPs. The output also includes latency and BW values for the fabric routing path for use in generating associated CDAT information.

Possible Command Return Codes:

- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Busy

Command Effects:

- Background Operation

Table 7-123. Get PID Binding Request Payload

Byte Offset	Length in Bytes	Description
0h	1	Target VCS : ID of the VCS to query.
1h	1	Target vPPB : Index of the vPPB to query. Reserved when the binding target is a Host ES VCS.

Table 7-124. Get PID Binding Response Payload

Byte Offset	Length in Bytes	Description
00h	2	- Bits[11:0]: PID: PID of the remote binding target. FFFh if unbound. - Bits[15:12]: Reserved.
02h	2	Reserved
04h	8	Latency Entry Base Unit : Latency Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Valid only when the binding target is a Downstream ES VCS.
0Ch	2	Latency Entry : Latency Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Valid only when the binding target is a Downstream ES VCS.
0Eh	8	BW Entry Base Unit : Bandwidth Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Valid only when the binding target is a Downstream ES VCS.
16h	2	BW Entry : Bandwidth Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Valid only when the binding target is a Downstream ES VCS.

7.7.13.7 Configure PID Binding (Opcode 5706h)

This command configures the binding of a PID to a target. It is used to bind:

- Downstream ES PIDs to Upstream ES vDSPs
- Upstream ES USP PIDs to Downstream ES vUSPs

The command input includes latency and BW values for the fabric routing path for use in generating associated CDAT information.

Possible Command Return Codes:

- Unsupported
- Background Command Started
- Invalid Input
- Internal Error
- Retry Required
- Busy

Command Effects:

- Background Operation

Table 7-125. Configure PID Binding Request Payload (Sheet 1 of 2)

Byte Offset	Length in Bytes	Description
00h	1	- Bits[2:0]: Operation: 000b = Bind 001b = Unbind - All other encodings are reserved - Bits[7:3]: Reserved
01h	1	Target VCS : ID of the VCS to which the PID is being bound.
02h	1	Target vPPB : Index of the vPPB to which the PID is being bound. Reserved when the binding target is a Downstream ES VCS.
03h	1	Reserved

Table 7-125. Configure PID Binding Request Payload (Sheet 2 of 2)

Byte Offset	Length in Bytes	Description
04h	2	- Bits[11:0]: PID: PID of the remote binding target - Bits[15:12]: Reserved
06h	2	Reserved
08h	8	Latency Entry Base Unit : Latency Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when the binding target is a Host ES VCS.
10h	2	Latency Entry : Latency Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when the binding target is a Host ES VCS.
12h	8	BW Entry Base Unit : Bandwidth Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when the binding target is a Host ES VCS.
1Ah	2	BW Entry : Bandwidth Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when the binding target is a Host ES VCS.

7.7.13.8 Get Table Descriptors (Opcode 5707h)

This command reads descriptors of the DPID Routing Tables and Routing Group Tables in a PBR Switch.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-126. Get Table Descriptors Request Payload

Byte Offset	Length in Bytes	Description
0h	2	Start Index : Starting index into list of descriptors.
2h	2	Number of Descriptors : Number of descriptors to read.

Table 7-127. Get Table Descriptors Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Start Index : Starting index into list of descriptors.
2h	2	Number of Descriptors : Number of table descriptors.
4h	Varies	Get Table Descriptors List : List of table descriptors as defined in Table 7-128.

Table 7-128. Get Table Descriptor Format

Byte Offset	Length in Bytes	Description
0h	1	- Bits[1:0]: Table Type: 00b = DRT 01b = RGT - All other encodings are reserved - Bits[7:2]: Reserved
1h	2	Table Index : Index of table.
3h	20h	Active Port Mask : Bitmask defining which ports actively use (1) or do not actively use (0) this table. Bit position corresponds to physical port number.
23h	4	Reserved

7.7.13.9 Get DRT (Opcode 5708h)

This command reads the DPID Routing Tables in a PBR Switch.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-129. Get DRT Request Payload

Byte Offset	Length in Bytes	Description
0h	1	DRT Index : Index of DRT to read.
1h	1	Reserved
2h	2	Number of Entries : Number of DRT entries to read.
4h	2	Start Entry : Starting index into DRT entries.

Table 7-130. Get DRT Response Payload

Byte Offset	Length in Bytes	Description
0h	1	DRT Index : Index of DRT.
1h	1	Reserved
2h	2	Number of Entries : Number of DRT entries.
4h	2	Start Entry : Starting index into DRT entries.
6h	1	Associated RGT Index : Index of RGT used by this DRT.
7h	1	Reserved
8h	Varies	DRT Entry List : List of DRT entry values as defined in Table 7-131 .

Table 7-131. DRT Entry Format

Byte Offset	Length in Bytes	Description
0h	1	- Bits[1:0]: Entry Type: Type of routing target specifier and M2S Req routing: 00b = Invalid 01b = Physical Port number 10b = RGT index 11b = Reserved - Bits[7:2]: Reserved
1h	1	Routing Target : Encoding depends on Entry Type : 00h = Reserved 01h = Physical port number 02h = RGT entry index

7.7.13.10 Set DRT (Opcode 5709h)

This command sets the DPID Routing Tables in a PBR Switch.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-132. Set DRT Request Payload

Byte Offset	Length in Bytes	Description
0h	1	DRT Index : Index of DRT to configure.
1h	1	Reserved
2h	2	Number of Entries : Number of DRT entries to configure.
4h	2	Start Entry : Starting index into DRT entries.
6h	Varies	DRT Entry List : List of DRT entry values as defined in Table 7-131.

7.7.13.11 Get RGT (Opcode 570Ah)

This command reads the Routing Group Tables in a PBR Switch.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-133. Get RGT Request Payload

Byte Offset	Length in Bytes	Description
0h	1	RGT Index: Index of RGT.
1h	1	Reserved
2h	2	Number of Entries: Number of RGT entries.
4h	2	Start Entry: Starting index into RGT entries.

Table 7-134. Get RGT Response Payload

Byte Offset	Length in Bytes	Description
0h	1	RGT Index: Index of RGT.
1h	1	Reserved
2h	2	Number of Entries: Number of RGT entries.
4h	2	Start Entry: Starting index into RGT entries.
6h	Varies	RGT Entry List: List of RGT entry values as defined in Table 7-135.

Table 7-135. RGT Entry Format

Byte Offset	Length in Bytes	Description
0h	1	Egress Port[0]: Physical port number.
1h	1	Egress Port[1]: Physical port number.
2h	1	Egress Port[2]: Physical port number.
3h	1	Egress Port[3]: Physical port number.
4h	1	Egress Port[4]: Physical port number.
5h	1	Egress Port[5]: Physical port number.
6h	1	Egress Port[6]: Physical port number.
7h	1	Egress Port[7]: Physical port number.
8h	1	• Bits[2:0]: Highest Valid Entry: Highest index in the Egress Port list that is valid. • Bits[5:3]: Highest Primary Entry: Highest index in the Egress Port list that specifies a primary routing path. Subsequent valid egress ports are considered secondary paths. • Bits[7:6]: Reserved.
9h	1	• Bits[2:0]: Dynamic Routing Mode: Specifies the dynamic routing mode to be used for this entry: — 000b = Random — 001b = Congestion Avoidance — 010b = Advanced Congestion Avoidance — 011b, 101b = Reserved — 110b, 111b = Vendor-specific • Bits[5:3]: Mix Setting: Specifies the mix used for dynamic routing mode, as defined in Section 7.7.6.3 • Bits[7:6]: Reserved
Ah	2	Reserved

7.7.13.12 Set RGT (Opcode 570Bh)

This command configures the Routing Group Tables in a PBR switch.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-136. Set RGT Request Payload

Byte Offset	Length in Bytes	Description
0h	1	RGT Index: Index of RGT to configure.
1h	1	Reserved
2h	2	Number of Entries: Number of RGT entries to configure.
4h	2	Start Entry: Starting index into RGT entries.
6h	Varies	RGT Entry List: List of RGT entry values as defined in Table 7-134 .

7.7.13.13 Get LDST/IDT Capabilities (Opcode 570Ch)

This command retrieves a vPPB's LDST and IDT Capabilities, per [Section 7.7.9](#).

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-137. Get LDST/IDT Capabilities Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .

Table 7-138. Get LDST/IDT Capabilities Response Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	2	Number of Segments: Number of LDST segments that are supported by this LDST/IDT. The number of entries must be 0 or a power of 2.
3h	1	LDST Segment Size - Bits[2:0]: LSegSz per the FSegSz encoding defined in Table 7-81 - Bits[7:3]: Reserved The device shall return 0h if this value has not been initialized.
4h	2	Number of IDT: Number of Interleave Device Table entries supported by this LDST/IDT.
6h	2	Number of Completer ID-Based Re-Routers: Number of Completer ID-Based Re-Router entries supported by this LDST/IDT.
8h	2	- Bits[11:0]: Local PID: PID assigned to this vPPB. FFFh if unassigned. - Bits[15:12]: Reserved.
Ah	8	Fabric Base: Base HPA of this LDST. FabricBase shall be aligned to the programmed LDST Segment Size. The device shall return 0h if this value has not been initialized.
12h	8	Fabric Limit: Upper HPA of this LDST. Shall be greater than FabricBase. Shall be aligned to the programmed LDST Segment Size. The device shall return 0h if this value has not been initialized.

7.7.13.14 Set LDST/IDT Configuration (Opcode 570Dh)

This command sets the GAE's LDST and IDT Capabilities, per [Section 7.7.9](#). Because the FabricBase and FabricLimit values must be aligned to the programmed LDST Segment Size, all three Host-chosen values are configured in one request.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-139. Set LDST/IDT Configuration Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	LDST Segment Size - Bits[2:0]: LSegSz per the FSegSz encoding defined in Table 7-81 - Bits[7:3]: Reserved
2h	8	FabricBase: Base HPA of this LDST. FabricBase shall be aligned to the programmed LDST Segment Size. The value 0h will disable this LDST/IDT decoder.
Ah	8	FabricLimit: Upper HPA of this LDST. Shall be greater than FabricBase. Shall be aligned to the programmed LDST Segment Size. The value 0h will disable this LDST/IDT decoder.

7.7.13.15 Get LDST Segment Entries (Opcode 570Eh)

This command reads the configuration of LDST Segment entries. The Host is responsible for mapping the LD-FAM range of HPAs to the appropriate number of available Segment Entries. Should the Host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-140. Get LDST Segment Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Seg Count: Number of LDST Segment Entries requested. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Segment Index: Index of the first segment being requested. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index given shall not be larger than the maximum segment entry number supported. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.

Table 7-141. Get LDST Segment Entries Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Seg Count: Number of LDST Segment Entries described in the Seg Entry_List[]. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
2h	2	Starting Segment Index: Index of the first segment being returned. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index given shall not be larger than the maximum segment entry number supported. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.
4h	Varies	Segment List[]: List of Segment Entries as defined in Table 7-142.

Table 7-142. LDST Segment Entry Format

Byte Offset	Length in Bytes	Description
0h	1	Valid - Bit[0]: Valid Entry: As per Figure 7-31 - Bit[1]: Enable PCIe Segment: Indicates that the target is in a separate PCIe segment, thus the request will include the requester's segment number - Bits[7:2]: Reserved
1h	1	Intlv - Bits[3:0]: Interleave Mode: As per Table 7-82 - Bits[7:4]: Reserved
2h	1	Gran - Bits[3:0]: Interleave Granularity: As per Table 7-83 - Bits[7:4]: Reserved
3h	1	Reserved
4h	2	DPID/IX: DPID or IDT Index, depending on Intlv field value: - Bits[11:0]: - If Intlv == 0, this is the actual DPID to which the LD-FAM request is to be sent. - Else, this is Index of the IDT entry that contains the DPID of the first EP in the interleave set. See Figure 7-31 and the description of interleaving in Section 7.7.2.4. - Bits[15:12]: Reserved
6h	2	Reserved

7.7.13.16 Set LDST Segment Entries (Opcode 570Fh)

This command is used by the Host to set the configuration of LDST Segment entries. The Host is responsible for mapping the LD-FAM range of HPAs to the appropriate number of available Segment Entries, per Section 7.7.2.4. Should the Host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

This command fails with Invalid Input if access to the specified DPID is not enabled in the LAV.

Possible Command Return Codes:

- Success
- Unsupported

- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-143. Set LDST Segment Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Seg Count: Number of LDST Segment Entries described in the Seg Entry_List[]. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Segment Index: Index of the first segment being configured. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index given shall not be larger than the maximum segment entry number supported. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.
6h	Varies	Segment List[]: List of Segment Entries as defined in Table 7-142 .

7.7.13.17 Get LDST IDT DPID Entries (Opcode 5710h)

This command reads the configuration of IDT entries that are used by the LDST. The Host is responsible for mapping the capacity of specific devices targeted by LDST into interleaved regions of HPA. Should the Host or the switch mailbox have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-144. Get LDST IDT DPID Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	LDST IDT Entry Count: Number of LDST IDT Entries requested. Value should be >0 and not more than the lesser of the total LDST IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target switch mailbox.
4h	2	Starting LDST IDT Entry Index: Index of the first LDST IDT entry being requested. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the LDST IDT Entry Count value shall not be larger than the maximum LDST IDT entry number supported.

Table 7-145. Get LDST IDT DPID Entries Response Payload

Byte Offset	Length in Bytes	Description
0h	2	LDST IDT Entry Count: Number of LDST IDT Entries returned. Value should be >0 and not more than the lesser of the total LDST IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target switch mailbox.
2h	2	Starting LDST IDT Entry Index: Index of the first LDST IDT entry being returned. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the LDST IDT Entry Count value shall not be larger than the maximum LDST IDT entry number supported.
4h	Varies	LDST IDT DPID[]: DPID of the target device for the LDST IDT entry. See Figure 7-31 and the description of interleaving in Section 7.7.2.4 . Repeats LDST IDT Entry Count number of times. - Bits[11:0]: PID of the target device - Bits[15:12]: Reserved

7.7.13.18 Set LDST IDT DPID Entries (Opcode 5711h)

This command sets the configuration of IDT entries that are used by the LDST. The Host is responsible for mapping the capacity of specific devices targeted by LDST into interleaved regions of HPA. Should the Host or the switch mailbox have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change